Hoapa Project Report

Notice: this document is the draft/wireframe for a talk, to present the Hoapa project, for an audience who are mostly logician, but which should be open to anyone with some philosophical and/or mathematical background. Do not modify it. What follows is the draft proposed by ig, and see 'Prompt' for what to do with it.

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0. Context and project history

- Pitch
- Opening remarks: mixtes, collective science
- The initial exercise
- Points of view adopted in this presentation
- Communication up to now & Zalamea
- The epistemological landscape
- Lautman turning point
- The argument
- A theory of everywhere (a generalization of Yoneda), of work, of in-betweens (mixtes)
- 1. As an implementation
- 2. The implementation in logic
- 3. The context of work
- 4. The dialectics of abstract work
- 5. The conjectural adjunction
- 6. Permacultural logic
- Conclusion: objective & future work

1. Hoapa as an implementation

- 1.1 Scale & context (by scale we mean "at the level of collective organization")
  - ...
- 1.2 Operations
  - ...
- 1.3 Biological metaphor
  - Energy = Open source
  - Metabolism = Hoapa
  - Neuron = Citizen assembly
  - Memory = Complex knowledge graphs (homotopy, etc.)
  - Internal language/logic = self-produced language (groupoidal layer)
  - Sociability = Open network layer ("federation")

2. The implementation in model and set theoretical logic

- ...

3. Thinking about abstract work

- 3.1 Work in the context of Kauffman (see arc/23)
- 3.2 Adding a loop into the abstract (Peirce, biosemiotics, Patttee, von Neumann, Deacon, Mossio & al., etc.)
- 3.3 Branching into Lautmanian dialectics: concrete & abstract

4. Hoapa in the dialectics of Lautman

- 4.1 Concrete / abstract (following the ideas in 3.)
- 4.2 Synthetic / analytic ()
- 4.3 Variation / individuation
- 4.4 Structure / existence | genesis

5. The conjectural adjunction

- 5.1 The adjunction in Zalamea's Lautman (where 1 = K, 2 = M, 3 = the category resulting from the adjunction):
    - Long version, in Structure and Unity:

> "{According Lautman, mathematical theories are constructed following 'a whole series of precisions, limitations, exceptions', through which the Ideas and concepts acquire their effective life. In fact, 'restrictions and delimitations [. . .] should not be conceived as an impoverishment, but rather as an enrichment of knowledge, due to the increased precision and certainty provided.'}[1] {Nevertheless, the exact, concrete and differentiated impetus consigned in this way has as a constant complement an inverse alternating impetus, oriented towards integrality and unity.}[2] In the science of the 1930s, the ineluctable emergence of the notion of mathematical structure gave Lautman a supple instrument to order the diversity. Still nascent when he was writing, and not yet defined in all their generality (the labour of Bourbaki), mathematical structures included a wide network that underlies the work of the German school (Hilbert, Emmy Noether, Emil Artin), studied closely by Lautman during his final years at the École Normale, with Herbrand, Chevalley and Ehresmann. {He was capable of grasping the emergence of various levels of mathematical structure: structured concrete objects (groups, algebras, Hilbert spaces, etc.), intermediate structural correspondences (duality theorems, Galois correspondences, conformal representations, etc.), generic structures (lattices, etc.). Going even further, he began to see 'structural schemas' in which – at an 'upper' logical level: more simple and universal – a number of free oppositions are assembled (local/global, intrinsic/extrinsic, discrete/continuous, etc.) throughout collections of possible structures.}[3]"

    - Short version, in Platonism:

> "What is proper and specific to mathematical activity is then to transit between a {world of ideas of possible, free relations}[2], and a {world much richer in determinations, full of precisions and delimitations}[1]. {The back-and-forth is inevitable}[3], and it opens up new possibilities for a Platonism better adapted to the

> complexities of mathematical reality: a structural and dynamic Platonism,
> not reified, immobile or eternal."

- ... (quick overview, methodology (use a metaphor where AI is the elements, sea and wind, enabling motion, Agda is the ship enabling persistence and safeguarding coherence, I and the anchoring authors provide the direction), expected result (see motivations))

6. Permacultural logic

- ... (*viral* logic takes from context, exploits, exhausts and doesn't allow continuation; *permacultural* logic does BWR; Nature uses both—see articles by Patrick Forterre, notably "To be or not to be alive"; I would like to invite the logicians in the audience to embark on the task on making it rigorous in pure logic, so as to give a rigorous base to an economy of degrowth based on voluntary work, and contrast it meaningfully with the exploitative-to-exhaustion viral/cancerous logic of forced labour capitalism)

Conclusion

... (objective of the talk: demonstrate that the work done and future work are serious enough to earn attention and effort, and pass on a message to any department of education which applies the "Finnish style/philosophy" that they should implement an embryonic version of the protocol: autonomous and open science workshops)

Prompt:

For every '-', please point to the most obvious, important, valuable and explanatory pieces of information that should be used, in the dried-out style explained in hoapa/AGENTS.md
Please fill out the parts marked with '...', again in the dried-out style.
For each, tell me in curly brackets if you want to comment on the draft/talk itself.
In the following sections you'll find my take on what I think should be there. Don't hesitate to add/remove/modify if you see fit.
Put your work into a file named communication/project_report_{name_of_agent}_{timestamp}.md

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Previous version of context, for reference:

The pitch

Start with how one can combine notions in the philosophy of mathematics with ideas from the engineering of networks, to think an extension of the scientific method to the collective which incorporates participative democracy, or more generally a theory of work*. And recognize both what it can do, and its place in Nature.

* For logicians, we can add: constrained by *permacultural* logic.

The context

I started my journey into epistemology as the concerned citizen I've always been, but by chance, when I stumbled upon an interesting question, or rather an exercise: what is the most universal and balanced communication system possible? (And how to make this question rigorous.)

I call this system, and the whole project of understanding it in depth, Hoapa.

In this open letter addressed to Fernando Zalamea, I will present the point of view on the construction emanating from those specifications which best exhibits how important it could be, both as a *product* of science, and to *itself*, the scientific activity.

While my communication around my work has focused on the direct action prescribed by the latter, in this text I will embrace my role in the development of the former. Essentially, I'm now ready for the "big book" and 10 years of solitary work to which Zalamea urged me to commit.

The turning point for me has been the comprehension of Albert Lautman, after many perplexing attempts. In particular the movement he sees between dialectical or metaphysical problems (which can justify the construction), and solutions to those found in effective mathematical or physical theories (which resemble the construction).

The first part of the letter thus considers Hoapa through a series of dualities: concrete & abstract, synthetic & analytic, individuality & variation, in a context regulated by a tension between structure and genesis.

In the second part we broaden the epistemological scope, and revisit those concepts through the different authors who also gave them life in their work: Alexander Grothendieck, Bill Lawvere, Gilbert Simondon, Stuart Kauffman, Gilles Deleuze, Charles S. Peirce. Using those I will reconstruct the state the project is currently in: a conjectural adjunction between Hoapa as a process and some category representing "nice mathematics".

Third I will translate the design of the construction in the language of the Helsinki Logic Group, where I will present my work in the fall of 2026. This adaptation into model and set theoretic logic should demonstrate that our seemingly esoteric considerations do produce interesting and useful structures for the working mathematician.

I will also try and convince this audience that the logic they know and love is implicitly *ecologist*, that its structural laws participate in a universe designed for sustainability and solidarity.

The name, Hoapa, comes from one of the many San languages and means "everywhere", as an homage to this culture, now largely extinct, who are the proud descendants of the first implementers of absolutely universal and balanced communication systems, masters of sustainability and solidarity.

Hoapa is too humble and respectful of difference to be a theory of everything, but I believe its network shape and resilience allows it to be a theory of everywhere, true in every context.

Ideas for:

- Opening remarks

In the 19th century mathematicians were doing analysis, or arithmetic, or geometry, etc. In the 20th century they started doing algebraic geometry, differential geometry, etc. We could say that there are domains of science that are mostly left unexplored (with the notable exception of mathematical-physics, in each possible pair the reductionist agenda wants to debase the second element and picture it as "just the first element"; for each example we put in parenthesis the exceptions we're most interested in): philosophical mathematics (championed by Zalamea, contra analytic philosophy of mathematics à la Russell/Wittgenstein), physical-biology (Kauffman, EES), biological-psychology (attachment theory), organizational-social-science

(Simondon, who says the psychological and social co-evolve, so the adequate first partner of the social is not the psychological but a more general "theory of organizations"). Hoapa's implementation lives in the last item in this list (and in the case of science is aimed at replacing the ad hoc system of peer reviews, mudded by its ties with the political history of "civilization"). But its theory is really an invariant in the general domain formed by all those mixtes/bridges.

- Epistemological landscape

Besides making Kauffman the entry point into the point of view of Hoapa as a "theory of work", made out of constraints specified by separations of concerns, themselves guided by the dialectics, and embedded in the "primary" concrete/abstract tension, satisfying the closure conditions of Kauffman. The newest addition to our series of anchors is Peirce (a repositioning rather), which we can now say is the epistemological anchor for this theory of work viewpoint.

- The argument

Abstract work can't reward its concrete substrate, the relation between abstract work and concrete substrate (where what 'abstract' and 'concrete' is is relative, as in some purely abstract things like mathematical objects can be considered concrete in relation to other mathematical objects which are even more abstract, as in Lautman) must be dedicated to a) mapping the context, in "intelligent" systems ([concrete <-> abstract] <-> world/context), b) the design (c -> a), and c) application (a -> c) of algorithms. The decomposition of abstract work in intelligent systems between a class of "synthetic (in the sense of Lautman) agents (as in the actor model)" and "analytic (Lautman) documented concepts (diagrammatic, many possible sources are relevant here—the analytic side is well developed in current science)", if we think of it as an object in a category of possible organizations of abstract work, would be a special object of that category (Hoapa's view of optimality).

- Theory of everywhere

See relevant section in com*/presentation*.pdf. To which I would like to add the idea that it is a generalization of the Yoneda lemma to "worlds"/contexts, which can each be "reconstructed" from their web of relations to their neighboring worlds/contexts. And how this can be made precise following the concept of work as in 3. & 4., and mixtes as in the opening remarks and Zalamea's Lautman.

Clarifications:

Questions

1. Scope of the output — inventory or dried talk? Two readings. (a) A content inventory: for each slot, the material that must be covered, as pointers for you to write from. (b) The talk itself, written dried-out, so the file is close to deliverable. These produce very different documents. I lean (a), since "point to the most obvious… pieces of information" reads like inventory, but the ... fills read more like (b).

2. Length budget. A talk's running time decides what survives. Forty-five minutes and twenty minutes are different talks, and §0 alone lists sixteen items before §1 begins. If you give me a minute count I'll mark what fits and what gets cut; otherwise I'll flag over-capacity but fill everything.

3. Do I reproduce the full skeleton? My assumption: yes — the complete structure with fills in place, so it reads as a finished wireframe rather than a list of patches.

4. Filename. project_report_claude_2026-10-23.md unless you want a different agent token or a time component. "claude" matches the anchors_concepts_claude.md convention.

5. Two sections are both numbered 5, and the §0 table of contents doesn't match the section titles — the TOC has 3. The context of work and 4. The dialectics of abstract work against actual headings 3. Thinking about abstract work and 4. Hoapa in the dialectics of Lautman, and the TOC's item 6 is the second §5. Do I renumber in my version, or preserve your numbering exactly and flag it in curly brackets?

6. §2 is the largest unknown and I can't scope it without you. "The implementation in model and set theoretical logic" — the formal work is cubical Agda, which is HoTT, not model or set theory. Is there existing material translating it for that audience, is this to be invented in this pass, or is §2 meant to be a plan for that translation? For a Helsinki logic audience this is the section that decides whether the talk lands, and it's currently a single ....

7. Which presentation PDF? §"Theory of everywhere" says see relevant section in com*/presentation*.pdf. There are two: presentation_of_hoapa_as_pure_form_of_communication.pdf and presentation_with_preliminary.pdf. Tell me which and I'll read it; I haven't yet.

8. Should I use arc/22 (hoapa_from_notions, 350k)? It looks directly on-point for §4 and you didn't name it. If yes I'll read it before filling §4.

9. Do my earlier review findings get applied silently, or flagged? Several passages I reviewed now sit in the "Previous version, for reference" block — the San paragraph's accuracy problems, essence versus structure & existence in Lautman, "a universe designed for sustainability", "close sibling to covering spaces" without the hedge, "some category representing nice mathematics" misstating M, and the recurrence argument undercutting itself by saying content recurs. My assumption is: write the corrected version into the fills, and note each correction in curly brackets so nothing changes without you seeing it.

10. Is the talk still ahead? The draft says "fall of 2026" and today is 23 October. If it has happened or is imminent, that changes what "future work" in the conclusion should say.

11. One flag I'd rather raise now than bury in a bracket. "A theory of everywhere (a generalization of Yoneda)" will be the most scrutinized phrase in the talk for this audience. Yoneda is a theorem about an object being determined by its maps from all others; "worlds reconstructible from their web of relations to neighbouring worlds" is a suggestive analogy whose precise form isn't available. Do you want it stated as an analogy, as a research target with its hypotheses named, or left as it stands and I flag the exposure?

Give me 1, 2, 6 and 9 and I can start; the rest I'll assume as above and mark my assumptions in the file.

❯ 1. - are more like a, ... are mre like b

1. Just fill in what's asked, I'll do the cutting down later
2. yes, reproduce

3. Put in the hours/minutes and give yourself a nickname based on what you've been working on.
4. fixed, change of titles is voluntary
5. It should be a translation, item by item, of the list of parts in 2., into the language and concepts of set and model theory. Do your best in a first pass, and we'll refine in subsequent
   passes.
6. pure form
7. I thought arc/22 was in your context already, you may reread, and in general go get the info you need from the project files, don't worry about usage, make yourself confortable.
8. yes, as you said
9. it's in 3 weeks
10. analogy worth investigating to make rigorous (but Hoapa doesn't have the resources in its current form to do that any time soon, we can be upfront about that).

In general I mostly want you to inventorize ideas, to not forget elements that have surfaced in our work.

Further Notes:

- Part 2 should be "the conjectured adjunction in model and set theoretical logic", following com_drafts/for_hgl, and replace part 5
- This part should be introduced by the remark of how Abramsky (whose contextuality I had already incorporated in my landscape) commented at the end of a talk (which I watched for this talk)